

Stochastic Analysis

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Summer 2026¹

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Chapter 1

Stochastic Processes in Continuous Time

1.1 Basic definitions

A *stochastic process* is a mathematical model for the occurrence, at each moment after some initial time, of a random phenomena. The randomness is captured by the introduction of a measurable space $(\Omega, \mathcal{F})$, the *sample space* on which a probability measure can be placed. A *stochastic process* is a family $X = (X_t)_{t \in I}$ of random variables on $(\Omega, \mathcal{F})$ taking values in measurable space $(S, \mathcal{S})$, the *state space*. Here, I a finite or infinite time interval, i.e. $I = [0, T]$, for some $T > 0$ or $I = [0, \infty)$. In this course, we shall only consider stochastic processes taking values in $\mathbb{R}$ or $\mathbb{R}^d$. That is, $X_t : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ or $X_t : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^d, \mathcal{B}^d)$ where $\mathcal{B}$ and $\mathcal{B}^d$ denote the Borel σ -fields on $\mathbb{R}$ and $\mathbb{R}^d$, respectively. The Borel σ -field on $[0, t]$ is denoted $\mathcal{B}_t$. If the index set is not relevant, we denote a stochastic process by (X_t) instead of $(X_t)_{t \in I}$. For each $\omega \in \Omega$ we call $X(\omega) := (X_t(\omega))_{t \in I}$ a *trajectory* or *sample path* of X . Stochastic processes can be equal in different ways, listed below.

Definition 1.1.1 Let $X = (X_t)_{t \in I}$ and $Y = (Y_t)_{t \in I}$ be stochastic processes defined on the same probability space.

- X and Y are called *indistinguishable* if

$$\mathbb{P}[X_t = Y_t \text{ for all } t \in I] = 1.$$

- X and Y are called *modifications* if

$$\mathbb{P}[X_t = Y_t] = 1 \quad \text{for all } t \in I.$$

The definitions of indistinguishable and modification automatically require the sets $\{\omega : X_t(\omega) = Y_t(\omega), t \in I\}$, respectively, $\{\omega : X_t(\omega) = Y_t(\omega)\}$ to be measurable, which does not need to be the case in general. Moreover, it does not make sense to ask whether X and Y are modifications or

indistinguishable unless they are defined on the same probability space. They do not have to be defined on the same space to have the same finite dimensional distributions.

Definition 1.1.2 *The stochastic processes X and Y have the same finite dimensional distributions if*

$$\mathbb{P}[X_{t_i} \in B_i \text{ for all } i = 1, \dots, n] = \mathbb{P}[Y_{t_i} \in B_i \text{ for all } i = 1, \dots, n]$$

for all $0 \leq t_1 \leq \dots \leq t_n$, all $n \in \mathbb{N}$ and all $B_i \in \mathcal{B}$ ($i = 1, \dots, n$).

Proposition 1.1.3 *i) If X and Y are indistinguishable, then they are modifications.*

ii) If X and Y are modifications, then they have the same finite dimensional distributions.

PROOF:

i) For any $t \in I$,

$$\mathbb{P}[X_t = Y_t] \geq \mathbb{P}[X_t = Y_t \text{ for all } t \in I] = 1.$$

ii) Let $N_t = \{X_t \neq Y_t\}$. Then $\mathbb{P}(N_t) = 0$ for all $t \in I$. For all $B \in \mathcal{B}^d$,

$$\begin{aligned} \mathbb{P}[(X_{t_1}, \dots, X_{t_d}) \in B] &= \mathbb{P}[(X_{t_1}, \dots, X_{t_d}) \in B \setminus N_1 \cup \dots \cup N_d] \\ &= \mathbb{P}[(Y_{t_1}, \dots, Y_{t_d}) \in B \setminus N_1 \cup \dots \cup N_d] \\ &= \mathbb{P}[(Y_{t_1}, \dots, Y_{t_d}) \in B] \end{aligned}$$

□

Exercise 1 *Prove that the converse implications do not hold in general.*

Let us recall that a function f is called left-continuous if $\lim_{t_n \uparrow t} f(t_n) = f(t)$ and right-continuous if $\lim_{t_n \downarrow t} f(t_n) = f(t)$. The next result shows that the distinction between modifications and indistinguishability is not necessary if the trajectories of a stochastic process are either all left-continuous or all right-continuous.

Proposition 1.1.4 *If X and Y are modifications and all trajectories are left-continuous (or all trajectories are right-continuous), then X and Y are indistinguishable.*

PROOF: Under the assumptions on the trajectories,

$$A := \{X_t = Y_t \text{ for all } t \in I\} = \{X_t = Y_t \text{ for all } t \in I \cap \mathbb{Q}\}.$$

In particular, $\{X_t = Y_t \text{ for all } t \in I \cap \mathbb{Q}\} \in \mathcal{F}$ and

$$\begin{aligned} \mathbb{P}(A) &= 1 - \mathbb{P}(X_t \neq Y_t \text{ for some } t \in I \cap \mathbb{Q}) \\ &\geq 1 - \sum_{t \in I \cap \mathbb{Q}} \mathbb{P}(X_t \neq Y_t) \\ &= 1. \end{aligned}$$

□

Strictly speaking, a stochastic process is a function of a pair of variables (ω, t) . It is thus reasonable to ask for some joint measurability properties.

Definition 1.1.5 A stochastic process X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called measurable, if the mapping $(t, \omega) \mapsto X_t(\omega)$ from $(I \times \Omega, \mathcal{B} \otimes \mathcal{F})$ to $(\mathbb{R}, \mathcal{B})$ is measurable.

It is an immediate consequence of Fubini's theorem that the trajectories of a measurable stochastic process are Borel-measurable functions of time and that the same holds for the function $t \mapsto \mathbb{E}X_t$ provided the components of X have defined expectations.

In order to model the flow of information, we introduce the notion of a filtration.

Definition 1.1.6 A family of σ -fields $\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$ ($s \leq t$) on $(\Omega, \mathcal{F})$ is called a filtration. We set $\mathcal{F}_\infty := \sigma(\cup_{t \geq 0} \mathcal{F}_t)$. A probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ equipped with a filtration is called a filtered probability space or stochastic basis.

The concept of measurability of a stochastic process introduced above is a weak one. The use of filtrations opens up the possibility of introducing more advanced concepts of measurability.

Definition 1.1.7 • A stochastic process X defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ is called

- progressively measurable with respect to $(\mathcal{F}_t)$ if for all $t \in I$ the mapping $(\omega, s) \mapsto X_s(\omega)$ from $(\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t)$ to $(\mathbb{R}, \mathcal{B})$ is measurable;
- adapted to the filtration $(\mathcal{F}_t)$ if the mapping $\omega \mapsto X_t(\omega)$ from $(\Omega, \mathcal{F}_t)$ to $(\mathbb{R}, \mathcal{B})$ is measurable for all $t \in I$.
- The progressive σ -field on $\Omega \times [0, \infty)$ generated by a filtration $(\mathcal{F}_t)$ is the smallest σ -field such that all progressive processes (considered as mappings on $\Omega \times [0, \infty)$) are measurable.
- The predictable σ -field on $\Omega \times [0, \infty)$ generated by a filtration $(\mathcal{F}_t)$ is the smallest σ -field such that all left-continuous adapted processes (considered as mappings on $\Omega \times [0, \infty)$) are measurable.

Exercise 2 (*)

i) Show that the progressive σ -field is the σ -field on $\Omega \times [0, \infty)$ is given by the set

$$\{A \subseteq \Omega \times [0, \infty) : A \cap (\Omega \times [0, t]) \in \mathcal{F}_t \otimes \mathcal{B}_t\}.$$

In other words, show that a process is progressively measurable if and only if it is measurable w.r.t. the progressive σ -field.

(ii) Show that the σ -fields generated by the following sets of functions coincide:

- a) the set of left-continuous functions
- b) the set of all elementary functions, that is functions of the form

$$X_t(\omega) = e_{-1}(\omega) + \sum_{j=0}^n e_j(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

where $0 = t_0 < t_1 < \dots < t_n$ and e_j is $\mathcal{F}_{t_j}$ -measurable and e_{-1} is $\mathcal{F}_0$ -measurable and bounded.

c) the space of predictable processes of the form: for $A \in \mathcal{F}_s$ with $s > 0$, respectively, $A \in \mathcal{F}_0$

$$X_r(\omega) = \mathbf{1}_{A \times (s, t]}(\omega, r), \quad X_r(\omega) = \mathbf{1}_{A \times 0}(\omega, r)$$

d) the space of continuous functions.

If we endow the set $\Omega \times [0, T]$ with the progressive σ -field $\mathcal{P}$, then a real-valued stochastic process X is $\mathcal{P} - \mathcal{B}$ -measurable if and only if it is progressively measurable in the sense of the above definition. Progressive measurability guarantees that time-integrals of stochastic processes are measurable and, as we will see later, suitably stopped processes have good measurability properties as well.

Exercise 3 Let X be a progressively measurable non-negative (or integrable) stochastic process. Prove that the process

$$Y_t(\omega) := \int_0^t X_s(\omega) ds$$

is progressively measurable. Hint: Use Tonelli's (Fubini's) theorem.

Let us now recall the following result from Analysis III. If $X_n : (\tilde{\Omega}, \tilde{\mathcal{F}}) \rightarrow (E, \mathcal{E})$ where (E, d) is a metric space, equipped with its Borel- σ -algebra $\mathcal{E}$, is measurable for all $n \in \mathbb{N}$ and

$$X_n(\omega) \rightarrow X(\omega)$$

for all $\omega \in \tilde{\Omega}$, then X is measurable as well. Indeed, let $A \subseteq E$ be closed and $A_n := \{x \in E : d(x, A) < 1/n\}$. Then A_n is open and

$$X^{-1}(A) = \bigcap_{n \in \mathbb{N}} \bigcup_{k=1}^{\infty} \bigcap_{m=k}^{\infty} X_m^{-1}(A_n) \in \tilde{\mathcal{F}}.$$

We notice that this result does not hold in general if E is not a metric space.

Lemma 1.1.8 Let X be a progressively measurable stochastic process. Then X is measurable and adapted.

PROOF: In order to show measurability, define for any $t \in I$ the process

$$X^t : (\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t) \rightarrow (\mathbb{R}, \mathcal{B}), \quad X^t(\omega, s) := X(\omega, s).$$

Then X^t is measurable by assumption. From this we will conclude that the process

$$\tilde{X}^t(s, \omega) := X_{s \wedge t}(\omega) : (\Omega \times [0, \infty), \mathcal{F}_t \otimes \mathcal{B}_t) \rightarrow (\mathbb{R}, \mathcal{B})$$

is measurable as well and hence also measurable w.r.t $\mathcal{F} \otimes \mathcal{B}$. Since

$$X_s(\omega) = \lim_{t \rightarrow \infty} X_{s \wedge t}(\omega)$$

we conclude that X is $\mathcal{F} \otimes \mathcal{B}$ -measurable as the limit of measurable functions. Using Fubini's theorem this also shows that $X_t : (\Omega, \mathcal{F}_t) \rightarrow (\mathbb{R}, \mathcal{B})$ is measurable, and hence that X is adapted. $\square$

Exercise 4 Show that a measurable process does not have to be progressively measurable and that an adapted process does not need to be measurable.

We now state a partial inverse of the above result without proof.

Proposition 1.1.9 If X is adapted and measurable, then it has a progressively measurable modification.

The preceding result will not be used in this course. We are only interested in stochastic processes that have continuous sample paths, for which a stronger result can easily be established.

Proposition 1.1.10 *If X is adapted and all trajectories are right-continuous (or all trajectories are left-continuous), then X is progressively measurable.*

PROOF: Let us assume that all trajectories are right-continuous. We need to prove that the mapping $(\omega, s) \mapsto X_s(\omega)$ from $(\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t)$ to $(\mathbb{R}, \mathcal{B})$ is measurable. To this end, we define (left-continuous!) processes

$$X_s^n(\omega) := X_{\frac{k-1}{2^n}t}(\omega), \quad \frac{k-1}{2^n}t < s \leq \frac{k}{2^n}t, \quad k = 1, \dots, 2^n.$$

Since X is adapted, this defines measurable mappings

$$F^n : (\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t) \rightarrow (\mathbb{R}, \mathcal{B})$$

whose limit as $n \rightarrow \infty$ is X , restricted to $(\Omega \times [0, t])$, due to the right-continuity of the sample paths. Hence, X is progressively measurable. $\square$

Definition 1.1.11 *A stochastic process X is called continuous, if all trajectories are continuous. It is called càdlàg - which stands for “continue à droite, limites à gauche” - (respectively, càglàd) if all trajectories are right (left)-continuous with left (right) limits.*

Exercise 5 (*) *Prove that a càdlàg function $f : [0, \infty) \rightarrow \mathbb{R}$ has at most countably many jumps.*

The most important example of a continuous stochastic process for us is Brownian motion.

Definition 1.1.12 *A stochastic process B on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called (standard) Brownian motion (BM) if $B_0 = 0$ and B has*

- *independent increments: for all $0 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq s < t < \infty$,*

$$B_t - B_s \perp (B_{s_1}, \dots, B_{s_n})$$

- *stationary increments: for all $0 \leq s < t < \infty$,*

$$B_t - B_s \sim N(0, t - s)$$

- *almost surely continuous sample paths*

The standard example of a càdlàg process is the Poisson process.

Definition 1.1.13 *A stochastic process N on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called Poisson process with parameter $\lambda > 0$ if*

- $N_0 = 0$
- N is a càdlàg process
- N has independent increments: for all $0 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq s < t < \infty$,

$$N_t - N_s \perp (N_{s_1}, \dots, N_{s_n})$$

- N has Poisson distributed increments: for all $0 \leq s < t < \infty$,

$$\mathbb{P}(N_t - N_s = k) = \frac{(\lambda(t-s))^k}{k!} e^{-\lambda(t-s)}$$

More generally, a stochastically continuous process X is called a Lévy process if $X_0 = 0$ and if X is a càdlàg process with stationary independent increments.

Our focus will be on Brownian motion and integrals with respect to Brownian motions. Brownian motion is continuous by definition; it will turn out that the stochastic integrals can be chosen to be continuous. Hence almost all processes considered in this course will be continuous.

1.2 σ -fields and stopping times

Let $(\Omega, \mathcal{G}, (\mathcal{G}_t), \mathbb{P})$ be a filtered probability space and let

$$\mathcal{N} := \{A \subset \Omega : \text{there exists } B \in \mathcal{G}, A \subset B, \mathbb{P}(B) = 0\}$$

be the set of all null sets. The completion of $\mathcal{G}$ is given by $\tilde{\mathcal{G}} := \mathcal{G} \vee \mathcal{N}$. We can extend the probability measure $\mathbb{P}$ on $(\Omega, \mathcal{G})$ to a probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \tilde{\mathcal{G}})$ by $\tilde{\mathbb{P}}$

$$\tilde{\mathbb{P}}(A) = \mathbb{P}(B) \quad \text{for } A \in \tilde{\mathcal{G}} \text{ such that } A \Delta B \in \mathcal{N}.$$

The probability space $(\Omega, \tilde{\mathcal{G}}, \tilde{\mathbb{P}})$ is called the completion of $(\Omega, \mathcal{G}, \mathbb{P})$. A filtration $(\mathcal{G}_t)$ is called complete if $\mathcal{N} \in \mathcal{G}_0$ and hence $\mathcal{N} \in \mathcal{G}_t$ for all $t \geq 0$. We call $(\mathcal{G}_t)$ right-continuous if

$$\mathcal{G}_t = \bigcap_{s>t} \mathcal{G}_s =: \mathcal{G}_{t+}.$$

Definition 1.2.1 We say that a filtration $(\mathcal{G}_t)$ satisfies the usual conditions if it is complete and right-continuous.

The usual conditions are almost always assumed as they provide additional regularity. For instance, we shall see that martingales w.r.t. filtrations that satisfy the usual conditions have right continuous modifications if (and only if) their expected values are right-continuous in the time variable. They also guarantee that first entrance times are stopping times and that optional and stopping times coincide (see below).

Definition 1.2.2 For a stochastic process X we denote by $\tilde{\mathcal{F}}_t^X := \sigma(X_s, 0 \leq s \leq t)$ the filtration generated by X and put

$$\tilde{\mathcal{F}}_\infty^X := \sigma(X_t : t \in I).$$

The completion of $(\tilde{\mathcal{F}}_t^X)$ is defined by $\mathcal{F}_t^X := \tilde{\mathcal{F}}_t^X \vee \mathcal{N}$ for all $t \in [0, \infty]$.

Exercise 6 Give an example that shows that $(\tilde{\mathcal{F}}_t^X)$ is not right-continuous in general (we will see that the completion of the Brownian filtration is right-continuous).

In what follows we put $\bar{I} = [0, \infty]$ if $I = [0, \infty)$ and $[0, T]$ else.

¹For two sets A and B the symmetric difference $A \Delta B$ is defined by $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

Definition 1.2.3 Let $(\mathcal{F}_t)$ be a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$.

- The mapping $\tau : \Omega \rightarrow \bar{I}$ is called a *stopping time* if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \in I$. We define

$$\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in I\}$$

- The mapping $\tau : \Omega \rightarrow \bar{I}$ is called an *optional time* if $\{\tau < t\} \in \mathcal{F}_t$ for all $t \in I$. We define

$$\mathcal{F}_{\tau+} := \{A \in \mathcal{F} : A \cap \{\tau < t\} \in \mathcal{F}_t \text{ for all } t \in I\}$$

Exercise 7 i) Show that every stopping is an optional time and that the two concepts coincide if the filtration is right-continuous.

ii) Show that for any stopping time τ the set $\mathcal{F}_\tau$ is a σ -field.

iii) Show that for any stopping time τ the set $\{\tau \leq s\}$ belongs to $\mathcal{F}_\tau$ for all $s \in I$; in particular, τ is $\mathcal{F}_\tau$ -measurable.

Lemma 1.2.4 Let $I = [0, \infty)$ and let $X : \Omega \rightarrow \mathbb{R}$ be right-(or left-)continuous and adapted. Let $\Gamma \subset \mathbb{R}$ be non-empty, and consider the first entrance time

$$\tau_\Gamma := \inf\{t \geq 0 : X_t \in \Gamma\} \quad (\inf \emptyset := \infty).$$

- If Γ is open, then τ_Γ is an optional time.
- If X is continuous and Γ is closed, then τ_Γ is a stopping time.

PROOF: Let Γ be open and X be right- (or left-) continuous. Then, by the right/left-continuity of X

$$\{\tau_\Gamma < t\} = \bigcup_{0 \leq s < t} \{X_s \in \Gamma\} = \bigcup_{s \in [0, t) \cap \mathbb{Q}} \{X_s \in \Gamma\},$$

the latter being a countable union. Since X is adapted and Γ is measurable, $\{X_s \in \Gamma\} \in \mathcal{F}_s \subseteq \mathcal{F}_t$. As a result, $\{\tau_\Gamma < t\} \in \mathcal{F}_t$.

Let us now assume that Γ is closed set $d(x, \Gamma) := \inf\{|x - y| : y \in \Gamma\}$. Then the mapping $x \mapsto d(x, \Gamma)$ is continuous, $\Gamma = \{x : d(x, \Gamma) = 0\}$ and $s \mapsto d(X_s(\omega), \Gamma)$ is continuous. Hence

$$\begin{aligned} \{\tau_\Gamma \leq t\} &= \{\omega \in \Omega : \inf_{s \leq t} d(X_s(\omega), \Gamma) = 0\} \\ &= \{\omega \in \Omega : \inf_{s \in \mathbb{Q}, s \leq t} d(X_s(\omega), \Gamma) = 0\} \\ &= \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \mathbb{Q}, s \leq t} \{\omega : d(X_s(\omega), \Gamma) < \frac{1}{n}\}. \end{aligned}$$

By the above

$$\bigcup_{s \in \mathbb{Q}, s \leq t} \{\omega : d(X_s(\omega), \Gamma) < \frac{1}{n}\} \in \mathcal{F}_t.$$

This shows that $\{\tau_\Gamma \leq t\} \in \mathcal{F}_t$. □

If Γ is open and X is right-continuous, then we know that the process spends a non-trivial amount of time in Γ once it entered the set and we can approximate the process at any such time from the right.

The situation is different if Γ is closed. In this case, let us assume that X is only right-continuous and only defined on $[0, t]$ for $t \notin \mathbb{Q}$. Then we can approximate $d(X_t(\omega), \Gamma)$ only from the left but not from the right; hence right-continuity of the function $s \mapsto d(X_s(\omega), \Gamma)$ is of little use here. Now, if the function $s \mapsto d(X_s(\omega), \Gamma)$ jumps to zero in t , then

$$\inf_{s \leq t} d(X_s(\omega), \Gamma) = 0 \neq \inf_{s \in \mathbb{Q}, s \leq t} d(X_s(\omega), \Gamma) = 0.$$

The following exercise shows how to overcome this problem.

Exercise 8 *Prove that the first entrance time to an open set is a stopping time if the filtration satisfies the usual conditions. Give an example that shows that the first entrance time to an open set is not necessarily a stopping time if the filtration does not satisfy the usual conditions.*

The following properties are more or less obvious, except perhaps the last two that can be proved as an exercise.

Lemma 1.2.5 *i) If τ is an optional time and $\theta \in \mathbb{R}_+$, then $\tau + \theta$ is optional.*

ii) The sum, minimum and maximum of stopping times is a stopping time.

iii) The sum of optional times is an optional time.

iv) The infimum, supremum, limit inferior and limit superior of a sequence of optional times is an optional time.

v) The supremum of a sequence of stopping times is a stopping time.

vi) If T and S are stopping times and $A \in \mathcal{F}_S$, then $A \cap \{S \leq T\} \in \mathcal{F}_T$. In particular, if $S \leq T$, then $\mathcal{F}_S \subset \mathcal{F}_T$.

vii) If T and S are stopping times, then $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$ and each of the events $\{T < S\}$, $\{S < T\}$, $\{T \leq S\}$, $\{S \leq T\}$ and $\{T = S\}$ belongs to $\mathcal{F}_T \cap \mathcal{F}_S$.

The following result shows the usefulness of progressive measurability.

Proposition 1.2.6 *Let X be a progressively measurable stochastic process and τ be a stopping time. Then the random variable*

$$X_\tau(\omega) := X_{\tau(\omega)}(\omega)$$

defined on $\{\tau < \infty\}$ (and only there!) is $\mathcal{F}_\tau$ -measurable. Moreover, the stopped process

$$X^\tau = (X_{\tau \wedge t})$$

is progressively measurable.

PROOF: We first need to show that for any $B \in \mathcal{B}$ and all $t \in I$

$$\{X_\tau \in B\} \cap \{\tau \leq t\} \in \mathcal{F}_t.$$

This event equals

$$\{X_{\tau \wedge t} \in B\} \cap \{\tau \leq t\} = \{X_t^\tau \in B\} \cap \{\tau \leq t\}.$$

Hence it is enough to prove progressive measurability of the stopped process. To this end, we notice that the mapping

$$(s, \omega) \mapsto X_s(\omega) \quad \text{from } ([0, t] \times \Omega, \mathcal{B}_t \otimes \mathcal{F}_t) \text{ to } (\mathbb{R}, \mathcal{B})$$

is measurable by assumption. The mapping

$$(s, \omega) \mapsto (\tau(\omega) \wedge s, \omega) \quad \text{from } ([0, t] \times \Omega, \mathcal{B}_t \otimes \mathcal{F}_t) \text{ to } ([0, t] \times \Omega, \mathcal{B}_t \otimes \mathcal{F}_t)$$

is measurable because τ is a stopping time. Hence, the mapping $(s, \omega) \mapsto X_{\tau(\omega) \wedge s}(\omega)$ is progressively measurable as the composition of suitably measurable mappings.

□

1.3 Martingales in continuous time

In this section we extend some well-known results from discrete-time martingale theory to continuous time. Throughout, we fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

Definition 1.3.1 *A stochastic process $M = (M_t)$ is called a martingale (with respect to $(\mathcal{F}_t)$) if*

- M is $(\mathcal{F}_t)$ -adapted
- $M_t \in L^1(\mathbb{P})$ for every $t \in I$
- $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

The process is called a sub- respectively, super martingale if the equality in the third item is replaced by $\geq$, respectively, $\leq$.

From the theory of conditional expectation it is easy to see that an integrable, $(\mathcal{F}_t)$ -adapted process M is a martingale if and only if $\int_A M_t d\mathbb{P} = \int_A M_s d\mathbb{P}$ for all $A \in \mathcal{F}_s$ and all $s \leq t$.

1.3.1 Fundamental inequalities and a first convergence result

This section establishes a series of fundamental inequalities for (sub)martingales in continuous time. The following results are well known from discrete-time martingale theory ($I = \mathbb{N}$).

Theorem 1.3.2 • *(Doob's maximal inequality) If $(M_n, \mathcal{F}_n)_{n \in \mathbb{N}}$ is a non-negative sub-martingale (or a martingale), then*

$$\lambda^p \mathbb{P}(\sup_{m \leq n} |M_m| \geq \lambda) \leq \mathbb{E}|M_n|^p.$$

If $M_n \in L^p$ for $p > 1$, then

$$\mathbb{E}[\sup_{m \leq n} |M_m|^p] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}|M_n|^p.$$

- *(Martingale convergence) Let $(M_n, \mathcal{F}_n)_{n \in \mathbb{N}}$ be an L^p bounded (super)martingale, i.e.*

$$\sup_n \mathbb{E}|M_n|^p < \infty.$$

Then there exists a random variable M_∞ such that

$$\mathbb{E}|M_\infty|^p < \infty \text{ and } M_t \rightarrow M_\infty \text{ a.s.}$$

If $p > 1$ we also have convergence in L^p (this is not true for $p = 1$).

- (Optional sampling) Let (M_n) be a uniformly integrable discrete time $(\mathcal{F}_n)$ -martingale, and let σ, τ be two stopping times. Then

$$\mathbb{E}(M_\tau | \mathcal{F}_\sigma) = M_{\tau \wedge \sigma} \quad \text{a.s.}$$

Analogous results hold for sub- and supermartingales.

We now show that the preceding results carry over to continuous time martingales under continuity conditions on the same paths.

Theorem 1.3.3 (Doob's maximal inequality) *Let M be a right-continuous martingale (or a non-negative sub-martingale) and put*

$$M_t^* := \sup_{0 \leq s \leq t} |M_s|.$$

Then, M_t^* is measurable and for any $\lambda \geq 0$ and $t \geq 0$

$$\lambda^p \mathbb{P}(M_t^* \geq \lambda) \leq \mathbb{E}|M_t|^p.$$

Furthermore, if $p > 1$, then

$$\mathbb{E}|M_t^*|^p \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}|M_t|^p.$$

PROOF: By right-continuity we can write

$$M_t^* = \sup_{s \in [0, t] \cap \mathbb{Q}} (|M_s| \vee |M_t|).$$

In particular, the running maximum is measurable as a countable supremum of measurable functions.

We put

$$S_n := \left\{ s_{nk} := \frac{t}{2^n} k : 0 \leq k \leq 2^n \right\}.$$

Then $t \in S_n$ and $u \leq t$ for all $u \in S_n$. We consider the discrete-time process

$$M_v^{n,*} := \sup_{u \in S_n, u \leq v} |M_u|, \quad v \in S_n.$$

By the discrete time version of Doob's maximal inequality,

$$\lambda^p \mathbb{P}(M_t^{n,*} \geq \lambda) \leq \mathbb{E}|M_t^n|^p = \mathbb{E}|M_t|^p$$

and

$$\mathbb{E}|M_t^{n,*}|^p \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}|M_t|^p.$$

By right-continuity $M_t^{n,*} \uparrow M_t^*$ and the result follows from Fatou's lemma. $\square$

Corollary 1.3.4 *If (M_t) is a right-continuous (non-negative sub-) martingale. Then almost every sample path $M(\omega)$ is bounded on any bounded time interval.*

Next, we prove the optional sampling theorem for continuous martingales. In particular, stopped martingales are martingales.

Theorem 1.3.5 (Doob's Optional Sampling Theorem) *Let (M_t) be a uniformly integrable right-continuous martingale (e.g. M is L^p -bounded for $p > 1$). Let τ, σ be finite stopping times. Then*

$$\mathbb{E}[M_\tau | \mathcal{F}_\sigma] = M_{\tau \wedge \sigma} \quad \text{and} \quad \mathbb{E}[M_\tau | \mathcal{F}_{\sigma+}] = M_{\tau \wedge \sigma} \quad \text{a.s.}$$

In particular, the stopped process $M^\tau = (M_{t \wedge \tau})$ is a martingale. Similar results hold for sub-martingales.

PROOF: Let M be a martingale. The proof uses discrete time optional sampling along with the fact that M can be approximated by discrete time martingales, due to its right-continuity. For any $n \in \mathbb{N}$, let

$$S_n := \left\{ s_{n,k} := \frac{k}{2^n} : 0 \leq k \leq n2^n \right\}.$$

Next, we introduce the discretised random times

$$\tau_n := \sum_{1 \leq k \leq n2^n} s_{n,k} \mathbf{1}_{(s_{n,k-1}, s_{n,k}]}(\tau) + \infty \cdot \mathbf{1}_{\{\tau > n\}}, \quad \sigma_n := \sum_{1 \leq k \leq n2^n} s_{n,k} \mathbf{1}_{(s_{n,k-1}, s_{n,k}]}(\sigma) + \infty \cdot \mathbf{1}_{\{\sigma > n\}}.$$

Then, for all $u \in S_n$,

$$\{\tau_n \leq u\} = \bigcup_{k: s_{n,k} \leq u} \{\tau \leq s_{n,k}\} \in \mathcal{F}_u.$$

Hence τ_n is an $(\mathcal{F}_u)_{u \in S_n}$ -stopping time and $\tau_n \downarrow \tau$ a.s. Analogous results hold for σ_n . Since (M_t) is uniformly integrable and right-continuous, we thus get from Fatou's lemma that

$$\begin{aligned} \mathbb{E}|M_{t \wedge \tau}| &= \mathbb{E} \left[\liminf_{n \rightarrow \infty} |M_{t \wedge \tau_n}| \right] \\ &\leq \liminf_{n \rightarrow \infty} \mathbb{E}|M_{t \wedge \tau_n}| \\ &\leq \mathbb{E}|M_t| < \infty. \end{aligned}$$

In particular, the stopped process is integrable. Using similar arguments as in the proof of Proposition 1.2.6 we can show that it is also adapted.

The martingale property of the stopped process thus follows from the main statement of the theorem, which we now prove. Again by discrete optional sampling (applied to $\{(M_u, \mathcal{F}_u) : u \in S_n \cup \{\infty\}\}$)

$$\int_A M_{\tau_n} d\mathbb{P} = \int_A M_{\tau_n \wedge \sigma_n} d\mathbb{P} \tag{1.1}$$

for all $A \in \mathcal{F}_{\sigma_n}$ and hence all $A \in \mathcal{F}_{\sigma+}$ (since $\sigma_n > \sigma$ on the set $\{\sigma < \infty\}$ implies $\mathcal{F}_{\sigma+} \subset \mathcal{F}_{\sigma_n}$). By right-continuity and the convergence of the stopping times to τ , respectively σ from above

$$M_{\tau_n \wedge \sigma_n} \rightarrow M_{\tau \wedge \sigma}, \quad M_{\tau_n} \rightarrow M_\tau \quad \text{a.s.}$$

Thus, if we were allowed to interchange the limit $n \rightarrow \infty$ and the integration on both sides of equation (1.1), the result would follow. By Vitali's convergence theorem interchanging limit and integration is possible if the integrands form families of uniformly integrable random variables, which we now show.

By the de la Vallée-Poussin theorem a family (g_t) of real-valued random variables is uniformly integrable if and only if there exists a nonnegative nondecreasing convex function $f : [0, \infty) \rightarrow [0, \infty)$ such that $\frac{f(t)}{t} \rightarrow \infty$ as $t \rightarrow \infty$ and

$$L := \sup_{t \in [0, \infty]} \mathbb{E}f(|g_t|) < \infty.$$

Since M is uniformly integrable by assumption, such a function exists for M . The convexity of f yields that $f(|M|)$ is a sub-martingale. Hence, by the discrete time stopping theorem

$$f(|M_{\tau_n \wedge N}|) \leq \mathbb{E}[f(|M_N|) | \mathcal{F}_{\tau_n \wedge N}]$$

for each $n, N \in \mathbb{N}$. Hence, using that $\mathbf{1}_{\{\tau_n \leq N\}} \in \mathcal{F}_{\tau_n \wedge N}$ we get for each $n, N \in \mathbb{N}$ that

$$\begin{aligned} \mathbb{E}[f(|M_{\tau_n}|) \mathbf{1}_{\{\tau_n \leq N\}}] &= \mathbb{E}[f(|M_{\tau_n \wedge N}|) \mathbf{1}_{\{\tau_n \leq N\}}] \\ &\leq \mathbb{E}[\mathbb{E}[f(|M_N|) | \mathcal{F}_{\tau_n \wedge N}] \mathbf{1}_{\{\tau_n \leq N\}}] \\ &\leq \mathbb{E}[f(|M_N|) \mathbf{1}_{\{\tau_n \leq N\}}] \\ &\leq \sup_t \mathbb{E}[f(|M_t|)] < \infty =: L \end{aligned}$$

where $L < \infty$ because M is uniformly integrable. Hence,

$$\mathbb{E}[f(|M_{\tau_n}|) \mathbf{1}_{\{\tau_n < \infty\}}] \leq L,$$

and, accounting for that fact that the event $\{\tau = \infty\}$ may have positive probability,

$$\mathbb{E}[f(|M_{\tau_n}|)] \leq 2L.$$

This implies uniform integrability of the family (M_{τ_n}) ; similar considerations apply to $(M_{\tau_n \wedge \sigma_n})$. $\square$

Remark 1.3.6 We emphasize that the last step of the above proof used the fact that the discrete martingale (M_n) is closed by a random variable M_∞ so that $M_{\tau_n} = M_\infty$ on $\{\tau_n = \infty\}$. We did not assume/use that the continuous-time martingale is closed for which we have no proof yet; see also Exercise 12.

Exercise 9 (*) Suppose that (X_t) is a right-continuous sub-martingale and that σ, τ are two bounded stopping times with $\sigma \leq \tau$ a.s. Then X_τ and X_σ are integrable and

$$\mathbb{E}[X_\tau | \mathcal{F}_\sigma] \geq X_\sigma.$$

The following result shows that the right continuity assumption of the filtration is in some sense without loss of generality.

Corollary 1.3.7 A right continuous $(\mathcal{F}_t)$ -martingale M is also an $(\mathcal{F}_{t+})$ -martingale.

PROOF: It is clear that M is $(\mathcal{F}_{t+})$ -adapted. To establish the martingale property, we apply the previous theorem to $0 \leq s < t < \infty$. $\square$

We close this section with the continuous time version of the martingale convergence theorem.

Theorem 1.3.8 (Martingale convergence) *Let M be a right-continuous L^p bounded (sub)martingale where $p > 1$. Then there exists a random variable M_∞ such that*

$$\mathbb{E}|M_\infty|^p < \infty \quad \text{and} \quad M_t \rightarrow M_\infty \text{ a.s. and in } L^p.$$

If $p = 1$, then almost sure convergence (but not convergence in L^1) holds if M is continuous.

PROOF: Since M is L^p -bounded, so is the discrete-time (sub)-martingale $\{(M_n, \mathcal{F}_n)\}_{n \in \mathbb{N}}$. If $p > 1$, the discrete martingale convergence theorem yields a random variable M_∞ such that

$$M_n \rightarrow M_\infty \quad \text{a.s. and in } L^p.$$

Moreover, by the continuous-time maximal inequality,

$$\begin{aligned} \mathbb{P}(\sup_{t \geq m} |M_t - M_m| > \lambda) &= \limsup_{l \rightarrow \infty, l \in \mathbb{N}} \mathbb{P}(\sup_{l \geq t \geq m} |M_t - M_m| > \lambda) \\ &\leq \lambda^{-p} \limsup_{l \rightarrow \infty, l \in \mathbb{N}} \mathbb{E}|M_l - M_m|^p \\ &\leq \varepsilon \end{aligned}$$

for all $m \in \mathbb{N}$ large enough using discrete-time L^p -convergence. This proves a.s. convergence of M_t to M_∞ as $t \rightarrow \infty$ as

$$\mathbb{P}(\sup_{t \geq m} |M_t - M_\infty| > \lambda) \leq \mathbb{P}(|M_m - M_\infty| > \lambda/2) + \mathbb{P}(\sup_{t \geq m} |M_t - M_m| > \lambda/2).$$

The proof of convergence in L^p is very similar.

Let us consider the case $p = 1$ and assume that M is continuous. Then

$$\tau_n := \inf\{t \geq 0 : |M_t| \geq n\}$$

is a stopping time by Lemma 1.2.4. Hence the stopped process M^{τ_n} is a bounded, hence L^p -bounded ($p > 1$) martingale (from optional sampling) and so it follows from the above that

$$M_t \rightarrow M_\infty \text{ a.s. on } \{\tau_n = \infty\}.$$

Hence we are finished if we can prove that

$$\mathbb{P}\left(\bigcup_n \{\tau_n = \infty\}\right) = 1.$$

To this end, let $b := \sup_{t \geq 0} \mathbb{E}|M_t|$. This quantity is finite by assumption. By Doob's maximal inequality

$$\mathbb{P}\left(\sup_{0 \leq t \leq T} |M_t| > \lambda\right) \leq \frac{b}{\lambda}.$$

By monotone convergence,

$$\mathbb{P}(\sup_{t \geq 0} |M_t| > \lambda) = \lim_{n \rightarrow \infty} \mathbb{P}(\sup_{0 \leq t \leq n} |M_t| > \lambda) \leq \frac{b}{\lambda}$$

and so

$$\mathbb{P}(\tau_n < \infty) \leq \mathbb{P}(\sup_{t \geq 0} |M_t| \geq n) \leq \frac{b}{n-1}.$$

Hence

$$\mathbb{P}(\tau_n = \infty) = 1 - \mathbb{P}(\tau_n < \infty) \rightarrow 1.$$

□

1.3.2 Upcrossing lemma and submartingale convergence

In what follows we establish a stronger convergence result for sub- and supermartingales. To this end, we first recall Doob's up-crossing lemma for discrete time super-martingales.

Lemma 1.3.9 (Discrete time up-crossing lemma) *Let $(M_t, \mathcal{F}_t)_{t \geq 0}$ be a super-martingale and let $F = \{0 \leq t_1 < t_2 < \dots < t_n\} \subset [0, \infty)$ be a finite set of time points. Let $U_F(\alpha, \beta, M)$ be the number of up-crossings of the interval $[\alpha, \beta]$ by the restricted path $(M_{t_i})_{i=1}^n$. Then*

$$\mathbb{E}[U_F(\alpha, \beta, M)] \leq \frac{\mathbb{E}(M_{t_n} - \alpha)^-}{\beta - \alpha} \leq \frac{\mathbb{E}(M_{t_n}^-) + \alpha^+}{\beta - \alpha}.$$

For the case of a sub-martingale

$$\mathbb{E}[U_F(\alpha, \beta, M)] \leq \frac{\mathbb{E}(M_{t_n} - \alpha)^+}{\beta - \alpha} \leq \frac{\mathbb{E}(M_{t_n}^+) + |\alpha|}{\beta - \alpha}.$$

For continuous time super-martingales we put for any $I \subset [0, \infty)$

$$U_I(\alpha, \beta, M) := \sup\{U_F(\alpha, \beta, M) : F \subset I \text{ finite}\}.$$

Proposition 1.3.10 (Continuous time up-crossing lemma) *Let $(M_t, \mathcal{F}_t)_{t \geq 0}$ be a super-martingale and let $I \subset [0, T]$. Then*

$$\mathbb{E}[U_{I \cap \mathbb{Q}}(\alpha, \beta, M)] \leq \sup_{t \in [0, T] \cap \mathbb{Q}} \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha}.$$

If M is right-continuous, then

$$\mathbb{E}[U_I(\alpha, \beta, M)] \leq \sup_{t \leq T} \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha}.$$

If M is a right-continuous submartingale, then

$$\mathbb{E}[U_I(\alpha, \beta, M)] \leq \sup_{t \leq T} \frac{\mathbb{E}(M_t - \alpha)^+}{\beta - \alpha} \leq \sup_{t \leq T} \frac{\mathbb{E}(M_t^+) + |\alpha|}{\beta - \alpha}.$$

PROOF: Let $\{F_n\}$ be an increasing sequence of finite sets whose union is $I \cap \mathbb{Q}$. Then each finite set $F \subset I \cap \mathbb{Q}$ belongs to some F_n . Hence

$$U_{I \cap \mathbb{Q}}(\alpha, \beta, M) = \sup_n U_{F_n}(\alpha, \beta, M).$$

By the discrete time up-crossing lemma we know that $U_{F_n}(\alpha, \beta, M)$ is measurable, hence $U_{I \cap \mathbb{Q}}(\alpha, \beta, M)$ is measurable. Since $\{U_{F_n}(\alpha, \beta, M)\}_{n \in \mathbb{N}}$ is an increasing sequence it follows from monotone convergence that

$$\mathbb{E}[U_{I \cap \mathbb{Q}}(\alpha, \beta, M)] = \sup_n \mathbb{E}[U_{F_n}(\alpha, \beta, M)].$$

Let s_n be the largest element in F_n . Since $(M_t - \alpha)^-$ is a sub-martingale² we get for any $t \in I \cap \mathbb{Q}$ with $t \geq s_n$ that (note that we do not assume that $s_n \in I \cap \mathbb{Q}$)

$$\mathbb{E}[U_{F_n}(\alpha, \beta, M)] \leq \frac{\mathbb{E}(M_{s_n} - \alpha)^-}{\beta - \alpha} \leq \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha} \leq \sup_{t \in I \cap \mathbb{Q}} \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha}.$$

²If X is a sub-martingale, then X^+ is a sub-martingale. Hence this follows from the fact that $X^- = (-X)^+$.

This proves the first assertion. The case of a right-continuous super/sub-martingale is left as an exercise. $\square$

Exercise 10 (*)

- a) Prove the up-crossing lemma for a right-continuous super-martingale.
- b) Conclude that almost every sample path $M(\omega)$ of a right-continuous super-martingale admits left-limits everywhere on $(0, \infty)$.

The following result is more or less immediate corollaries of the upcrossing lemma (for right-continuous) submartingales. The proof of the first result is left as an exercise.

Proposition 1.3.11 (Submartingale convergence) *Let $(M_t, \mathcal{F}_t)_{t \geq 0}$ be a right-continuous submartingale that satisfies $\sup_{t \geq 0} \mathbb{E}(M_t^+) < \infty$. Then there exists an integrable random variable M_∞ such that*

$$M_n \rightarrow M_\infty \quad \mathbb{P}\text{-a.s.}$$

Exercise 11 (*) *Prove the above proposition.*

Hint. Use upcrossing on the time interval $[0, n]$, then monotone convergence and show that $\sup_{t \geq 0} \mathbb{E}(M_t^+) < \infty$ is equivalent to the seemingly stronger condition $\sup_t \mathbb{E}|M_t| < \infty$.

The following two exercises are very useful; the results should be known from Stochastic II. The first exercise requires Vitali's theorem.

Exercise 12 (*) *Prove that the following three conditions are equivalent for a non-negative right-continuous sub-martingale (M_t) .*

- a) (M_t) is a uniformly integrable family of random variables
- b) (M_t) converges in L^1 as $t \rightarrow \infty$
- c) (M_t) converges a.s. to some $M_\infty \in L^1$ and $(M_t)_{0 \leq t \leq \infty}$ is a sub-martingale

Exercise 13 (*) *Prove that the following four conditions are equivalent for a right-continuous martingale (M_t) .*

- a) (M_t) is a uniformly integrable family of random variables
- b) (M_t) converges in L^1 as $t \rightarrow \infty$
- c) (M_t) converges a.s. to some $M_\infty \in L^1$ and $(M_t)_{0 \leq t \leq \infty}$ is a martingale
- d) there exists $Y \in L^1$ such that $M_t = \mathbb{E}[Y | \mathcal{F}_t]$ a.s., for every $t \geq 0$.

Moreover, if d) holds, and M_∞ is the random variable from c), then $\mathbb{E}[Y | \mathcal{F}_\infty] = M_\infty$.

So far we have seen that right-continuity of (super-) martingales is a very useful property when it comes to convergence. The following section establishes sufficient conditions for the existence of right-continuous modifications.

1.3.3 Regularization and right-continuous modifications

We close this section with a useful result on the path regularities of (sub/super-)martingales. This result is the main reason why one typically assumes that a filtration satisfies the usual conditions.

Theorem 1.3.12 *Let M be an $(\mathcal{F}_t)$ -super-martingale and assume that $(\mathcal{F}_t)$ satisfies the usual conditions. Then M has a right continuous modification if and only if the function $t \mapsto \mathbb{E}M_t$ is right continuous. If the right continuous modification exists it is also a super-martingale and can be chosen to have left limits everywhere.*

The proof requires the following result. It shows that any super-martingale almost surely admits right and left limits everywhere.

Proposition 1.3.13 *Let M be a super-martingale. Then there exists a set $\Omega^* \in \mathcal{F}_\infty$ of measure one such that for all $\omega \in \Omega^*$ the following limits exist and are finite for all $t \in [0, \infty)$:*

$$\lim_{s \downarrow t, s \in \mathbb{Q}} M_s(\omega), \quad \lim_{s \uparrow t, s \in \mathbb{Q}} M_s(\omega).$$

PROOF: By the up-crossing lemma we know that the following event has measure zero for every $n \in \mathbb{N}$:

$$A_{\alpha, \beta}^n := \{\omega \in \Omega : U_{[0, n] \cap \mathbb{Q}}(\alpha, \beta, M(\omega)) = \infty\}.$$

The same is true for the event

$$A^n := \bigcup_{\alpha < \beta, \alpha, \beta \in \mathbb{Q}} A_{\alpha, \beta}^n.$$

This event contains an event that contains the event

$$\left\{ \omega \in \Omega : \liminf_{s \uparrow t, s \in \mathbb{Q}} M_s(\omega) < \limsup_{s \uparrow t, s \in \mathbb{Q}} M_s(\omega), \text{ for some } t \in [0, n] \right\}.$$

As a result, the left limit exists for all $t \geq 0$ and all $\omega \notin \bigcup_n A^n$. Similar arguments apply to the right limits.

We still need to prove that the limits are finite, i.e. that the sample paths are bounded on compact time intervals. To this end, let $t \in [0, \infty)$ and $n > t$. Let $\{Q_{n, m}\}_{m \in \mathbb{N}}$ be a sequence of finite sets of rational numbers that increase to $Q_n := [0, n] \cap \mathbb{Q}$, all containing 0 and n . Then $(M_s)_{s \in Q_{n, m}}$ is a discrete time super-martingale. Hence (assuming w.l.o.g. that it starts in 0),

$$\mathbb{P}\left(\max_{s \in Q_{n, m}} |M_s| > \lambda\right) \leq \frac{\mathbb{E}|M_n|}{\lambda}.$$

By the monotone convergence theorem for sets this shows that $\mathbb{P}(\sup_{s \in Q_n} |M_s| < \infty) = 1$. $\square$

For a given super-martingale M we now define $M_{t+}(\omega) := \lim_{s \downarrow t, s \in \mathbb{Q}} M_s(\omega)$ if this limit exists and is finite and zero otherwise. The process (M_{t+}) satisfies the following properties.

Lemma 1.3.14 *Let M be a super-martingale.*

- *The process (M_{t+}) is integrable and $\mathbb{E}[M_{t+} | \mathcal{F}_t] \leq M_t$ a.s. If the mapping $t \mapsto \mathbb{E}M_t$ is right-continuous, then this inequality is an equality.*
- *The process (M_{t+}) is an $(\mathcal{F}_{t+})$ -super-martingale and a martingale, if M is a martingale.*

Review: Backward martingales

The proof of the preceding lemma uses the following result on backward martingales that is known from Stochastic II.

Lemma 1.3.15 (Backwards martingale convergence) *Let (n_k) be a sequence of decreasing numbers and $(\mathcal{G}_{n_k})$ be a decreasing sequence of σ -fields and*

$$\mathcal{G}_\infty := \bigcap \mathcal{G}_{n_k}.$$

Let (X_{n_k}) be a $(\mathcal{G}_{n_k})$ -backwards super-martingale, that is

$$X_{n_{k+1}} \geq \mathbb{E}[X_{n_k} | \mathcal{G}_{n_{k+1}}].$$

Then the following holds:

- *The process (X_{n_k}) converges almost surely to a limit X_∞ .*
- *The process (X_{n_k}) is uniformly integrable if and only if it is L^1 bounded. In this case*

$$X_{n_k} \rightarrow X_\infty \text{ a.s. and in } L^1 \text{ and } X_\infty \geq \mathbb{E}[X_{n_0} | \mathcal{G}_\infty].$$

- *If X is a backwards martingale, then uniform integrability is automatically satisfied and*

$$X_\infty = \mathbb{E}[X_{n_0} | \mathcal{G}_\infty].$$

The uniform integrability of a backwards super-martingale (X_{n_k}) is equivalent to its L^1 -boundedness. Since

$$\mathbb{E}[|X_{n_k}|] = \mathbb{E}[X_{n_k}] + 2\mathbb{E}[X_{n_k}^-] \leq \mathbb{E}[X_{n_0}] + 2\mathbb{E}[X_{n_k}^-]$$

the latter follows if $\sup_k \mathbb{E}[X_{n_k}^-] < \infty$. To verify this condition, the following lemma is often useful.

Lemma 1.3.16 *Let $(M_t)_{t \geq 0}$ be a super-martingale with respect to a filtration $(\mathcal{F}_t)$. Then for every $0 \leq t \leq s$,*

$$\mathbb{E}[M_s^-] \leq \mathbb{E}[M_t^-].$$

PROOF: Fix $t \leq s$. For $K > 0$, we define the truncated variable

$$\phi_K(x) := (-x) \wedge K = \max(-x, -K) \cdot (-1)$$

so that $\phi_K(x) \uparrow x^-$ as $K \rightarrow \infty$. The mapping $x \mapsto \phi_K(x)$ is convex and of linear growth. Hence conditional expectations of $\phi_K(M_s)$ are well defined and the conditional Jensen inequality for convex functions gives

$$\mathbb{E}[\phi_K(M_s)] \leq \mathbb{E}[\phi_K(M_t)].$$

Since $\phi_K(M_t) \uparrow M_t^-$ pointwise as $K \rightarrow \infty$, and similarly $\phi_K(M_s) \uparrow M_s^-$ monotone convergence yields

$$\mathbb{E}[M_s^-] = \lim_{K \rightarrow \infty} \mathbb{E}[\phi_K(M_s)] \leq \lim_{K \rightarrow \infty} \mathbb{E}[\phi_K(M_t)] = \mathbb{E}[M_t^-].$$

□

The following corollary is key to the proof of Lemma [1.3.14](#)

Corollary 1.3.17 *Let M be a super-martingale and (q_n) be a decreasing sequence of real numbers that converge to t from above. Then the backwards super-martingale (M_{q_n}) is uniformly integrable.*

PROOF: It is enough to prove that $\sup_n \mathbb{E}[M_{q_n}^-] < \infty$. Since $q_n \geq t$ for all $n \in \mathbb{N}$ this follows from the previous lemma as

$$\mathbb{E}[M_{q_n}^-] \leq \mathbb{E}[M_t^-] < \infty.$$

□

Remaining Proofs.

Proof of Lemma 1.3.14. Let $q_n \downarrow t$ be a decreasing sequence of rational numbers. Then (M_{q_n}) is a backwards super-martingale. By Corollary 1.3.17 it is uniformly integrable. Hence the limit M_{t+} is integrable and $M_{q_n} \rightarrow M_{t+}$ in L^1 . Uniform integrability also allows to interchange limit and integration to obtain from $\mathbb{E}[M_{q_n}|\mathcal{F}_t] \leq M_t$ that

$$\mathbb{E}[M_{t+}|\mathcal{F}_t] \leq M_t, \quad \text{a.s.}$$

Moreover, L^1 convergence implies convergence of the expected values. Hence, $\mathbb{E}M_{q_n} \rightarrow \mathbb{E}M_{t+}$. Hence, if the mapping $t \mapsto \mathbb{E}M_t$ is right-continuous, then $\mathbb{E}M_{t+} = \lim_{n \rightarrow \infty} \mathbb{E}M_{q_n} = \mathbb{E}M_t$. Along with the previous inequality this implies that $\mathbb{E}[M_{t+}|\mathcal{F}_t] = M_t$ (cf. Exercise 14 below applied to $Y = M_t, X = M_{t+}$ and $\mathcal{G} = \mathcal{F}_t$).

To prove the second statement, let $s < t$ and let $\{q_n\}$ be a sequence of rational numbers decreasing to s . W.l.o.g. $q_n \leq t$. By the tower property of conditional expectations

$$\mathbb{E}[M_{t+}|\mathcal{F}_{q_n}] = \mathbb{E}[\mathbb{E}[M_{t+}|\mathcal{F}_t]|\mathcal{F}_{q_n}] \leq \mathbb{E}[M_t|\mathcal{F}_{q_n}] \leq M_{q_n}$$

with equality if M is a martingale. The right side converges to M_{s+} by definition; the left side converges to $\mathbb{E}[M_{t+}|\mathcal{F}_{s+}]$ by the backwards martingale convergence theorem. □

Exercise 14 *Let X, Y be integrable random variables on $(\Omega, \mathcal{F})$. Let $\mathcal{G} \subset \mathcal{F}$ be another σ -field. Let Y be $\mathcal{G}$ -measurable. Show that $\mathbb{E}[X|\mathcal{G}] \leq Y$ together with $\mathbb{E}X = \mathbb{E}Y$ implies $\mathbb{E}[X|\mathcal{G}] = Y$ a.s.*

We are now ready to prove the main result of this section.

PROOF OF THEOREM 1.3.12. By Proposition 1.3.13 and because the filtration is complete there exists a set $\Omega^* \in \mathcal{F}_0(!)$ of full measure on which the limits $M_{t\pm}$ exist for every $t \in [0, \infty)$. We set $\tilde{M}_t(\omega) = M_{t+}(\omega)$ on Ω^* and $\tilde{M}_t(\omega) = 0$ otherwise. Since M_{t+} is $\mathcal{F}_{t+}$ -measurable, $\tilde{M}_t$ is $\mathcal{F}_{t+}$ -measurable as well because the two variables differ only on a set of measure zero and the filtration is complete.

In view of the right-continuity of the filtration ($\mathcal{F}_t = \mathcal{F}_{t+}$), and it follows that $\tilde{M}_t$ is actually $\mathcal{F}_t$ -measurable because

$$\tilde{M}_t = M_{t+} = \mathbb{E}[M_{t+}|\mathcal{F}_{t+}] = \mathbb{E}[M_{t+}|\mathcal{F}_t].$$

Now, if the process of expected values is right-continuous, then Lemma 1.3.14 yields that $\tilde{M}_t = M_t$ a.s. for all $t \geq 0$. Hence $\tilde{M}$ is a modification of M .

To see the super-martingale property we take $s < t$, a set $A \in \mathcal{F}_s$ and a sequence of rational numbers $\{q_n\}$ that decreases to s ; w.l.o.g. $q_n < t$ for all $n \in \mathbb{N}$. From the super-martingale property of M and uniform integrability of the backwards super-martingale (M_{q_n}) we have that

$$\int_A \tilde{M}_t d\mathbb{P} = \int_A M_t d\mathbb{P} \leq \int_A M_{q_n} d\mathbb{P} \rightarrow \int_A \tilde{M}_s d\mathbb{P}$$

and so $\tilde{M}$ is a super-martingale. It follows from Exercise [10](#) that $\tilde{M}$ has left limits everywhere on a set of full measure. The converse implication is left as an exercise. $\square$

Exercise 15 *Prove missing converse implication in the proof of Theorem [1.3.12](#).*

Hint. Use backwards martingale convergence along with $|x| = 2(-x)\mathbf{1}_{\{-x>0\}} + x$.

